

1. General Description

The EM74HC245; EM74HCT245 are 8-bit transceivers with 3-state outputs. The device features an output enable ($\overline{OE}$) and send/receive (DIR) for direction control. A HIGH on $\overline{OE}$ causes the outputs to assume a high-impedance OFF-state. Inputs include clamp diodes. This enables the use of current limiting resistors to interface inputs to voltages in excess of V_{CC} .

2. Features and Benefits

- Wide supply voltage range from 2.0 V to 6.0 V
- High noise immunity
- CMOS low power dissipation
- Latch-up performance exceeds 250 mA
- Octal bidirectional bus interface
- Non-inverting 3-state outputs
- Complies with JEDEC standards:
 - JESD8C (2.7 V to 3.6 V)
 - JESD7A (2.0 V to 6.0 V)
- Input levels:
 - For EM74HC245: CMOS level
 - For EM74HCT245: TTL level
- ESD protection:
 - HBM ANSI/ESDA/JEDEC JS-001 Class 3A exceeds 5000 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options

EM74HC245; EM74HCT245

Octal buffer/line driver; 3-state

3. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EM74HC245D	HC245□ XYYWW	SOP-20L	plastic small outline package; 20 leads; body width 7.5 mm	2000
EM74HCT245D	HCT245□ XYYWW			
EM74HC245PW	HC245□ XYYWW	TSSOP-20L	plastic thin shrink small outline package; 20 leads; body width 4.4 mm	3000
EM74HCT245PW	HCT245□ XYYWW			

MARKING INFORMATION

NOTE: HC245□ = Serial number and Internal code; XYYWW = Vendor Code and Date Code.

HC245 □

Internal Code
Serial number

X YY WW

Date Code -Week
Date Code -Year
Vendor Code

4. Function Diagram

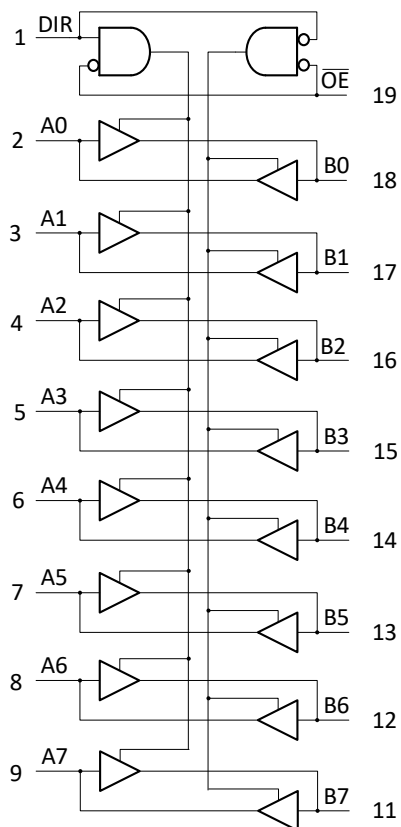


Fig. 1. Logic symbol

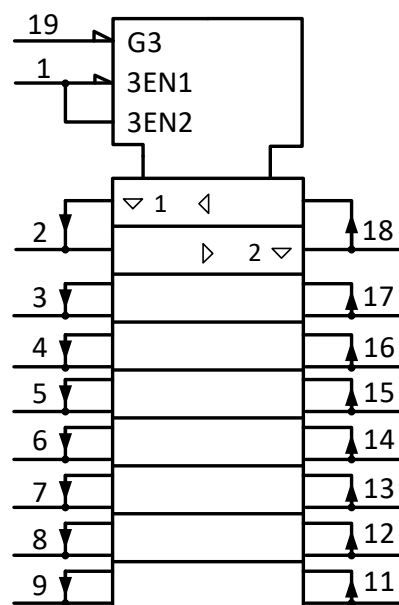
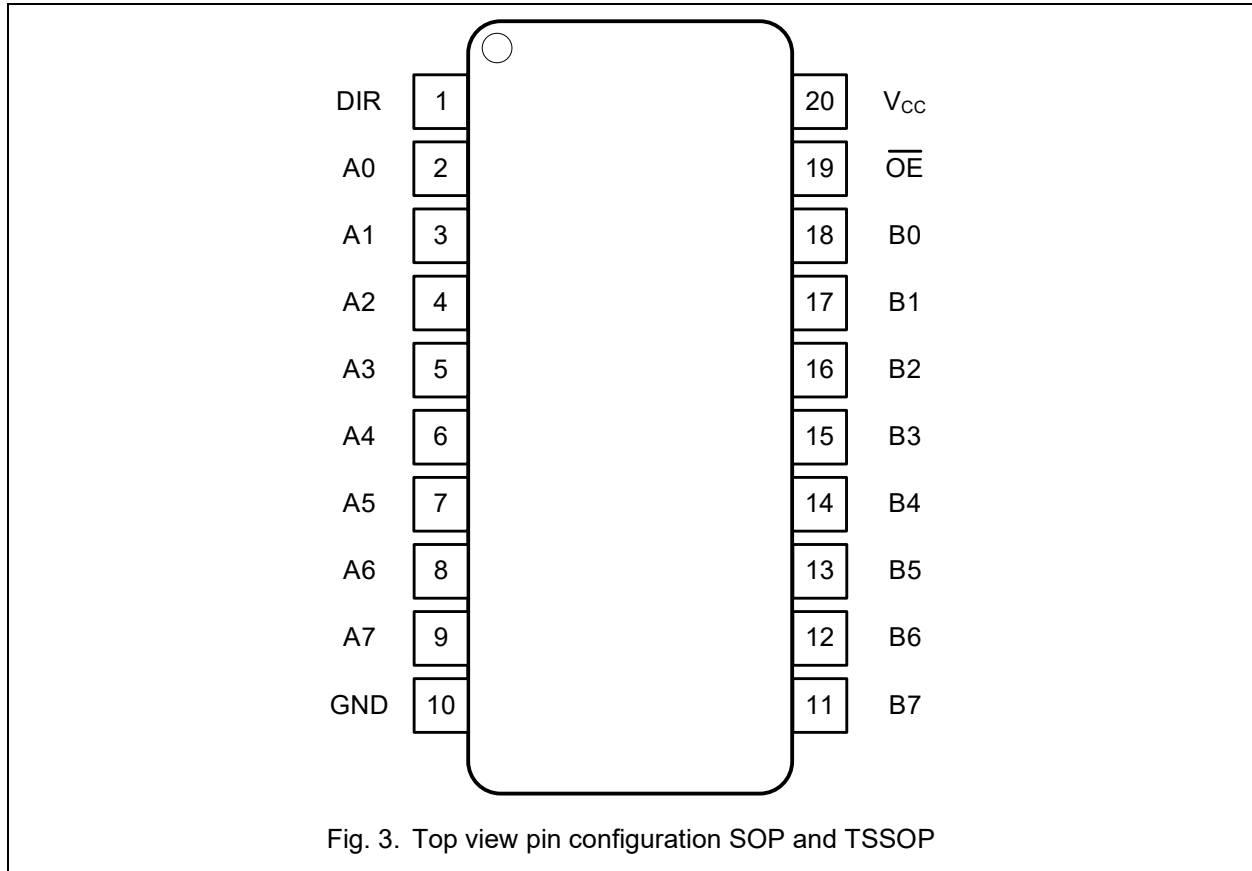


Fig. 2. IEC logic symbol

5. Pinning Information

5.1. Pinning



5.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
DIR	1	Direction control
A0, A1, A2, A3, A4, A5, A6, A7	2, 3, 4, 5, 6, 7, 8, 9	Data input/output
GND	10	Ground (0 V)
B7, B6, B5, B4, B3, B2, B1, B0	11, 12, 13, 14, 15, 16, 17, 18	Data input/output
$\overline{OE}$	19	Output enable input (active LOW)
V _{CC}	20	Supply voltage

6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level; X = don't care; Z = high-impedance OFF-state.

Input		Input/output	
$\overline{OE}$	DIR	An	Bn
L	L	A = B	input
L	H	input	B = A
H	X	Z	Z

7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	supply voltage		-0.5	7.0	V
I_{IK}	input clamping current	$V_I < -0.5\text{ V}$ or $V_I > V_{CC} + 0.5\text{ V}$ [1]		± 20	mA
I_{OK}	output clamping current	$V_O < -0.5\text{ V}$ or $V_O > V_{CC} + 0.5\text{ V}$ [1]		± 20	mA
I_O	output current	$V_O = -0.5\text{ V}$ to $(V_{CC} + 0.5\text{ V})$		± 25	mA
I_{CC}	supply current			50	mA
I_{GND}	ground current		-50		mA
P_{tot}	total power dissipation			500	mW
T_{stg}	storage temperature		-65	150	°C

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Voltages are referenced to GND.

Symbol	Parameter	Conditions	EM74HC245			EM74HCT245			Unit
			Min	Typ	Max	Min	Typ	Max	
V _{CC}	supply voltage		2.0	5.0	6.0	2.7	5.0	5.5	V
V _I	input voltage		0		V _{CC}	0		V _{CC}	V
V _O	output voltage		0		V _{CC}	0		V _{CC}	V
T _{amb}	ambient temperature		-40		125	-40		125	°C
$\Delta t/\Delta V$	input transition rise and fall rate	V _{CC} = 2.0 V			625				ns/V
		V _{CC} = 4.5 V		1.67	139		1.67	139	ns/V
		V _{CC} = 6.0 V			83				ns/V

EM74HC245; EM74HCT245

Octal buffer/line driver; 3-state

9.Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V). Typical values measured at $T_{amb} = 25^{\circ}\text{C}$ (unless otherwise noted).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
EM74HC245								
V _{IH}	HIGH-level input voltage	V _{CC} = 2.0 V	1.5			1.5		V
		V _{CC} = 4.5 V	3.15			3.15		V
		V _{CC} = 6.0 V	4.2			4.2		V
V _{IL}	LOW-level input voltage	V _{CC} = 2.0 V			0.5		0.5	V
		V _{CC} = 4.5 V			1.35		1.35	V
		V _{CC} = 6.0 V			1.8		1.8	V
V _{OH}	HIGH-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = -20 μA; V _{CC} = 2.0 V	1.9			1.9		V
		I _O = -20 μA; V _{CC} = 4.5 V	4.4			4.4		V
		I _O = -20 μA; V _{CC} = 6.0 V	5.9			5.9		V
		I _O = -6.0 mA; V _{CC} = 4.5 V	3.84			3.7		V
		I _O = -7.8 mA; V _{CC} = 6.0 V	5.34			5.2		V
V _{OL}	LOW-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = 20 μA; V _{CC} = 2.0 V			0.1		0.1	V
		I _O = 20 μA; V _{CC} = 4.5 V			0.1		0.1	V
		I _O = 20 μA; V _{CC} = 6.0 V			0.1		0.1	V
		I _O = 6.0 mA; V _{CC} = 4.5 V			0.33		0.4	V
		I _O = 7.8 mA; V _{CC} = 6.0 V			0.33		0.4	V
I _I	input leakage current	V _I = V _{CC} or GND ; V _{CC} = 6.0 V			±1		±1	μA
I _{OZ}	OFF-state output current	V _I = V _{IH} or V _{IL} ; V _{CC} = 6.0 V ; V _O = V _{CC} or GND			±5		±10	μA
I _{CC}	supply current	V _I = V _{CC} or GND ; I _O = 0 A ; V _{CC} = 6.0 V			20		40	μA
C _I	input capacitance			4.0				pF
C _{I/O}	Input/output capacitance			7.7				pF

EM74HC245; EM74HCT245

Octal buffer/line driver; 3-state

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
EM74HCT245								
V _{IH}	HIGH-level input voltage	V _{CC} = 3 V	2.0			2.0		V
		V _{CC} = 4.5 V	2.0			2.0		V
		V _{CC} = 5.5 V	2.1			2.1		V
V _{IL}	LOW-level input voltage	V _{CC} = 3 V			0.8		0.8	V
		V _{CC} = 4.5 V			0.8		0.8	V
		V _{CC} = 5.5 V			0.8		0.8	V
V _{OH}	HIGH-level output voltage	V _I = V _{IH} or V _{IL} ;						
		I _O = -20 μA; V _{CC} = 3 V	2.9			2.9		V
		I _O = -20 μA; V _{CC} = 4.5 V	4.4			4.4		V
		I _O = -6.0 mA; V _{CC} = 4.5 V	3.84			3.7		V
V _{OL}	LOW-level output voltage	V _I = V _{IH} or V _{IL} ;						
		I _O = 20 μA; V _{CC} = 3 V			0.1		0.1	V
		I _O = 20 μA; V _{CC} = 4.5 V			0.1		0.1	V
		I _O = 6.0 mA; V _{CC} = 4.5 V			0.33		0.4	V
I _I	input leakage current	V _I = V _{CC} or GND ; V _{CC} = 5.5 V			±1		±1	μA
I _{OZ}	OFF-state output current	V _I = V _{IH} or V _{IL} ; V _{CC} = 5.5 V ; V _O = V _{CC} or GND			±5		±10	μA
I _{CC}	supply current	V _I = V _{CC} or GND; I _O = 0 A; V _{CC} = 5.5 V			20		40	μA
ΔI _{CC}	additional supply current	per pin ; V _I = V _{CC} - 2.1 V; I _O = 0 A; other inputs at V _{CC} or GND; V _{CC} = 4.5 V to 5.5 V			400		490	μA
C _I	input capacitance			4.0				pF
C _{I/O}	Input/output capacitance			7.7				pF

10. Dynamic Characteristics

Table 7. Dynamic characteristics

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 6. Typical values measured at $T_{amb} = 25^{\circ}\text{C}$ (unless otherwise noted).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
EM74HC245								
t _{pd}	propagation delay	An to Bn or Bn to An; see Fig. 4 [1]						
		V _{CC} = 2.0 V			35		40	ns
		V _{CC} = 4.5 V			20		25	ns
		V _{CC} = 6.0 V			15		20	ns
t _{en}	enable time	$\overline{OE}$ to An or Bn; see Fig. 5 [2]						
		V _{CC} = 2.0 V			35		40	ns
		V _{CC} = 4.5 V			20		25	ns
		V _{CC} = 6.0 V			15		20	ns
t _{dis}	disable time	$\overline{OE}$ to An or Bn; see Fig. 5 [3]						
		V _{CC} = 2.0 V			35		40	ns
		V _{CC} = 4.5 V			20		25	ns
		V _{CC} = 6.0 V			15		20	ns
t _t	transition time	see Fig. 4 [4]						
		V _{CC} = 2.0 V			9		11	ns
		V _{CC} = 4.5 V			6		8	ns
		V _{CC} = 6.0 V			4		5	ns
C _{PD}	power dissipation capacitance	per buffer; V _I = GND to V _{CC} [5]		18				pF

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Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
EM74HCT245								
t _{pd}	propagation delay	An to Bn or Bn to An; V _{CC} = 4.5V; see Fig. 4 [1]			20		25	ns
t _{en}	enable time	$\overline{OE}$ to An or Bn; V _{CC} = 4.5V; see Fig. 5 [2]			20		25	ns
t _{dis}	disable time	$\overline{OE}$ to An or Bn; V _{CC} = 4.5V see Fig. 5 [3]			20		25	ns
t _t	transition time	V _{CC} =4.5V; see Fig. 4; [4]			6		8	ns
C _{PD}	power dissipation capacitance	per buffer; V _I = GND to V _{CC} – 1.5 V [5]		18				pF

[1] t_{pd} is the same as t_{PHL} and t_{PLH} .

[2] t_{en} is the same as t_{PZH} and t_{PZL} .

[3] t_{dis} is the same as t_{PLZ} and t_{PHZ} .

[4] t_t is the same as t_{THL} and t_{TLH} .

[5] C_{PD} is used to determine the dynamic power dissipation (P_D in μW).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \Sigma(C_L \times V_{CC}^2 \times f_o)$ where:

f_i = input frequency in MHz;

f_o = output frequency in MHz;

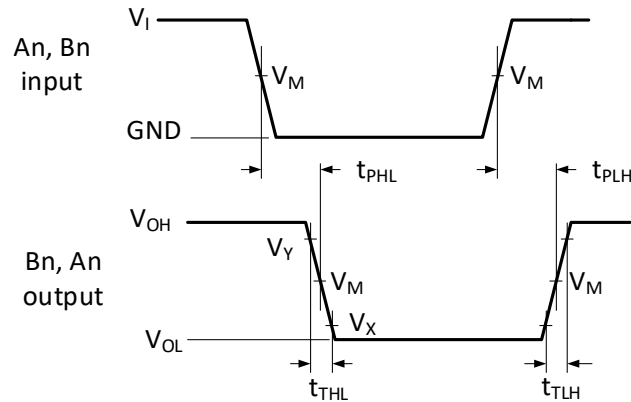
C_L = output load capacitance in pF;

V_{CC} = supply voltage in V;

N = number of inputs switching;

$\Sigma(C_L \times V_{CC}^2 \times f_o)$ = sum of outputs.

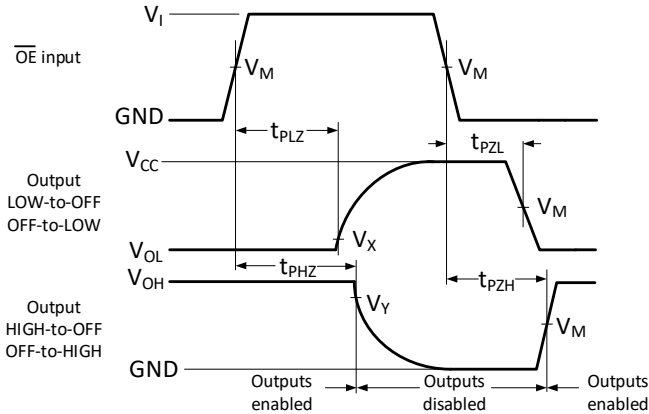
10.1. Waveforms and test circuit



Measurement points are given in Table 8.

V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

Fig. 4. Input (An, Bn) to output (Bn, An) propagation delays and output transition times



Measurement points are given in Table 8.

V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

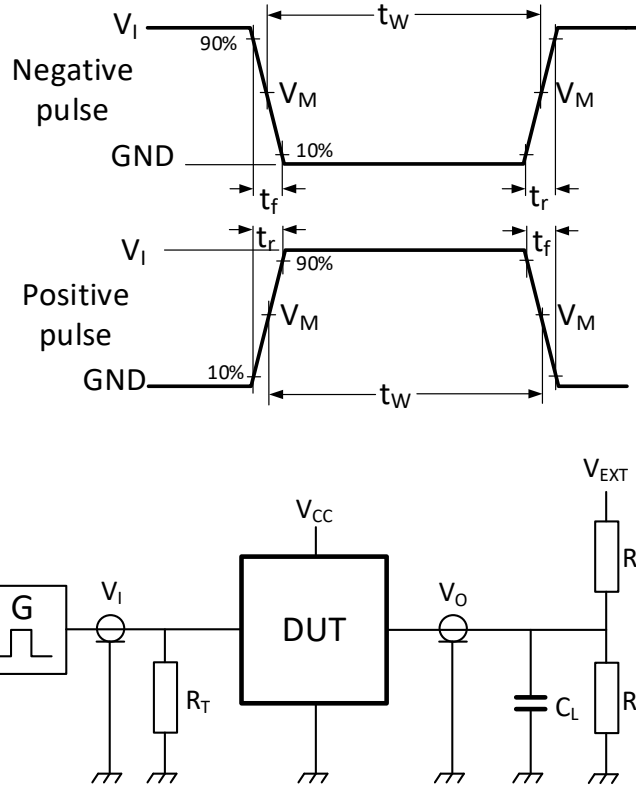
Fig. 5. 3-state enable and disable times

Table 8. Measurement points

Type	Input	Output		
	V_M	V_M	V_X	V_Y
EM74HC245	$0.5V_{CC}$	$0.5V_{CC}$	$0.1V_{CC}$	$0.9V_{CC}$
EM74HCT245	1.3 V	1.3 V	$0.1V_{CC}$	$0.9V_{CC}$

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Octal buffer/line driver; 3-state



Test data is given in Table 9.

Definitions for test circuit:

R_L = Load resistance.

C_L = Load capacitance including jig and probe capacitance.

R_T = Termination resistance should be equal to the output impedance Z_o of the pulse generator.

V_{EXT} = External voltage for measuring switching times.

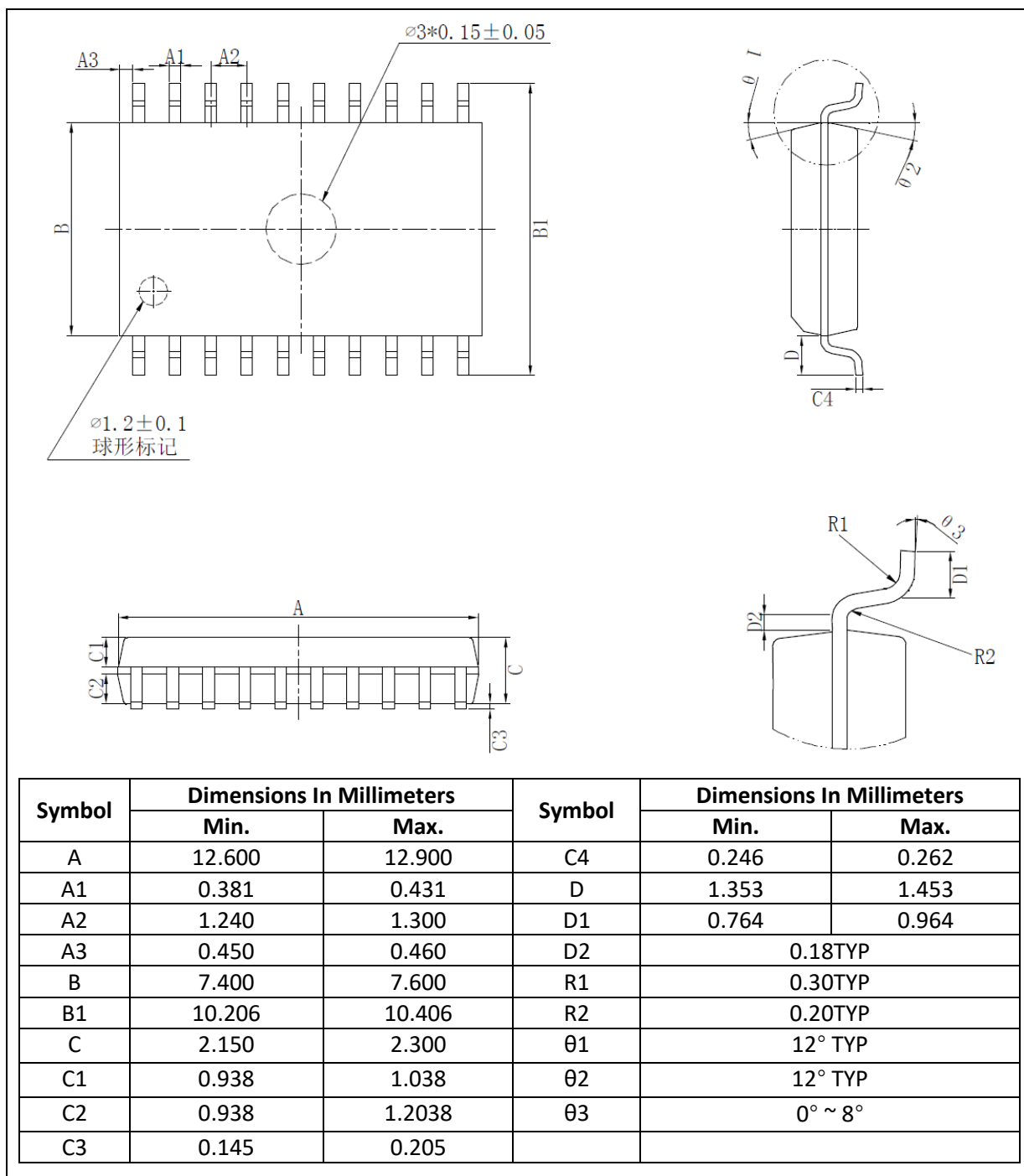
Fig. 6. Test circuit for measuring switching times

Table 9. Test data

Type	Input		Load		V_{EXT}		
	V_I	$t_r = t_f$	C_L	R_L	t_{PLH}, t_{PHL}	t_{PZH}, t_{PHZ}	t_{PZL}, t_{PLZ}
EM74HC245	V_{CC}	$\leq 2.5 \text{ ns}$	50 pF	500 Ω	open	GND	$2V_{CC}$
EM74HCT245	3 V	$\leq 2.5 \text{ ns}$	50 pF	500 Ω	open	GND	$2V_{CC}$

11. Package Outline

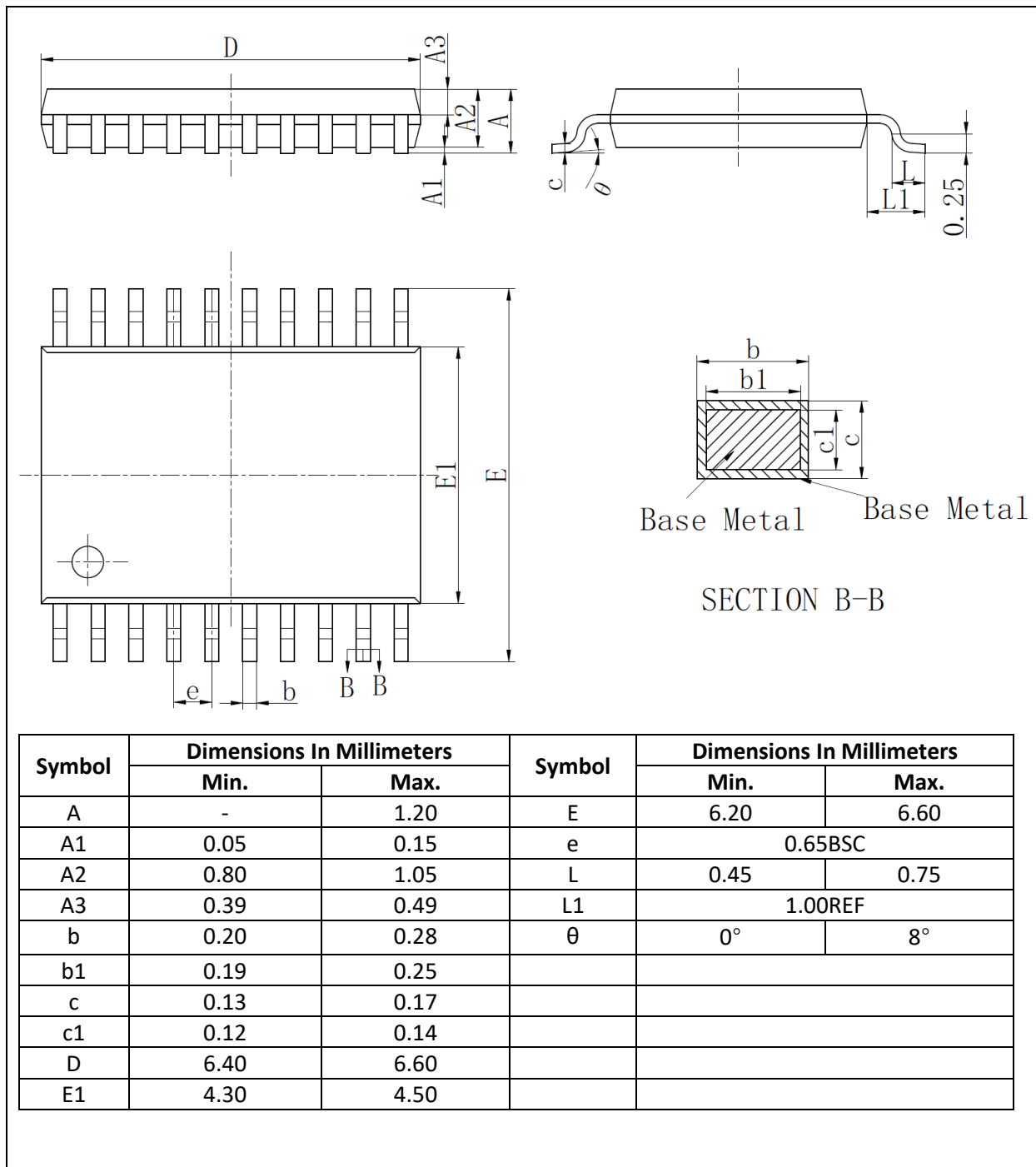
SOP-20L



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TSSOP-20L



12. Tape and Reel Information

12.1. Carrier tape dimensions

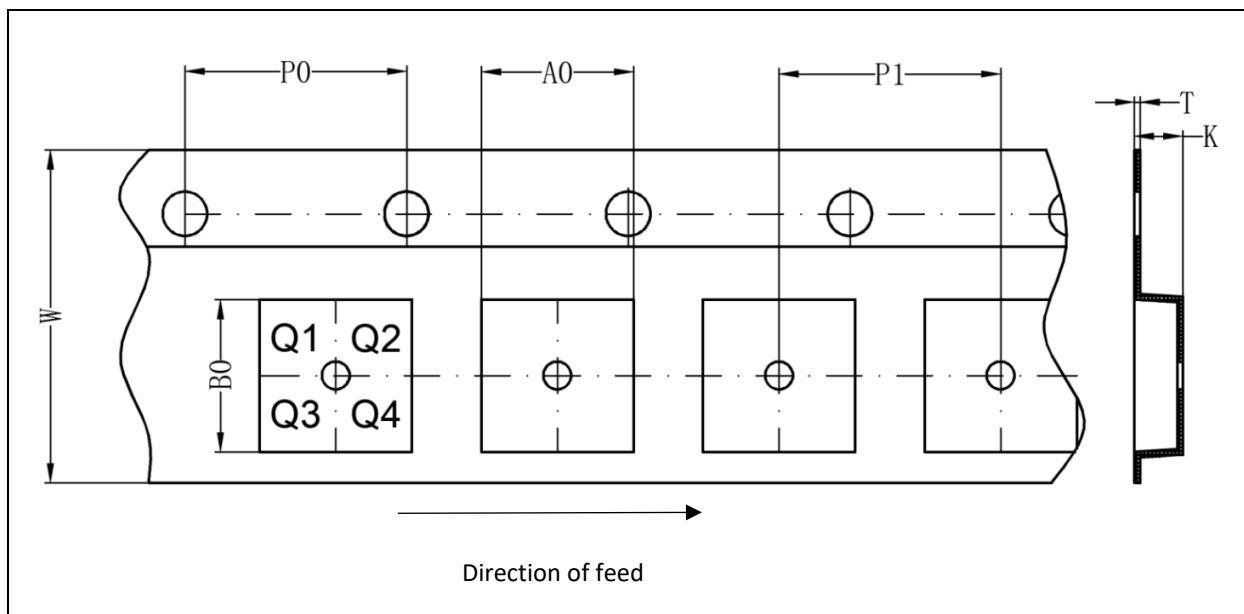


Table 10. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K0(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
SOP-20L	10.65	13.2	3.2	0.3	12	24	4	Q1
TSSOP-20L	6.85	6.85	1.7	0.22	8	12	4	Q1

EM74HC245; EM74HCT245

Octal buffer/line driver; 3-state

12.2. Reel and box dimensions

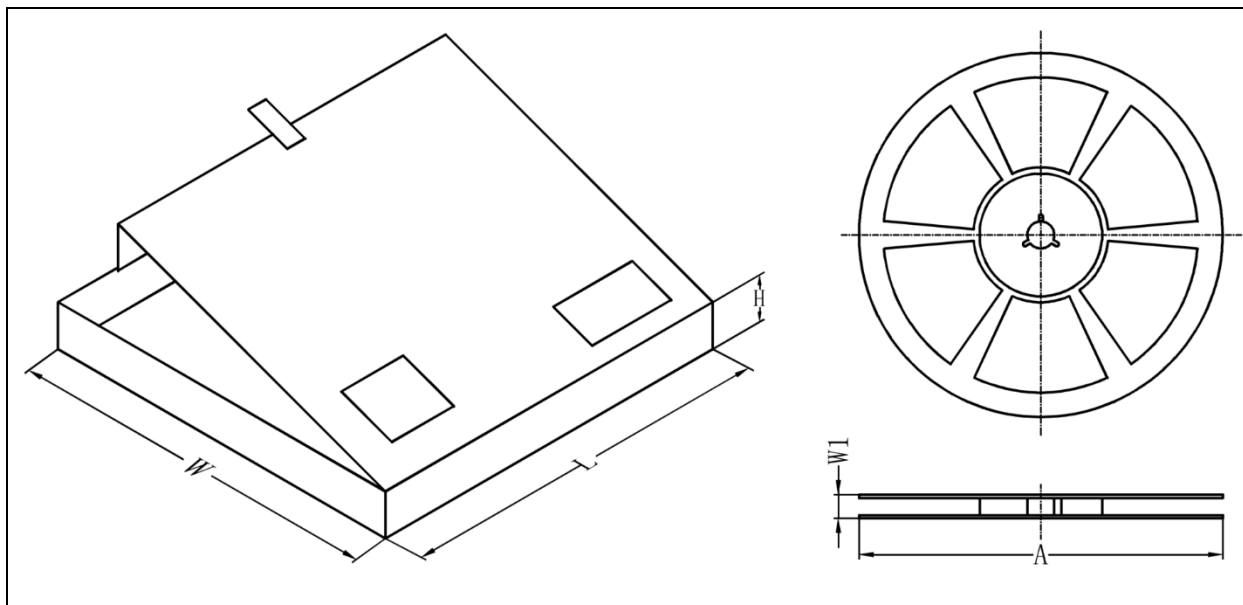


Table 11. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Reel Width W1 (mm)	SPQ (pcs) [1]	Reels per box	Outer box dimensions L×W×H(mm) [2]
SOP-20L	D	330	30.4	2000	1	358x340x50
TSSOP-20L	PW	330	18.4	3000	1	358x340x50

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

13. Abbreviations

Table 12. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	ElectroStatic Discharge
HBM	Human Body Model
CDM	Charged Device Model

14. Revision History

Table 13. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74HC_HCT245 Rev. 1.1	Dec 15, 2025	Product datasheet		EM74HC_HCT245 Rev. 1.0
Modifications:	- Table 1: topside marking information updated. - Section12: Added tape and reel information.			
EM74HC_HCT245 Rev. 1.0	Apr 18, 2025	Product datasheet		